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RESEARCH ARTICLE

# VesselTransGAN to CT Imaging: A Contrast Medium Free CTA Solution

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**ABSTRACT** Computed Tomography Angiography (CTA) is vital for detecting and planning treatment for vascular disease, but its use of contrast medium limits viability for patients with renal insufficiency. Based on the standard Computed Tomography (CT) scans, which did not require injected contrast medium, the cross-modal conversion technology generates CT-CTA images that provide reference properties equivalent to CTA diagnosis. Our research presented VesselTransGAN, a framework designed to create CTA from CT scans by leveraging the Generative Adversarial Network (GAN) framework and 2D-3D fusion strategy. This innovative approach comprised two key elements. Firstly, it employed pixel grayscale alignment to enhance blood vessels locally. Secondly, a 2D-3D fusion strategy improved each slice's image quality and continuity among slices. Through quantitative evaluation of image quality, compared with SOTA models, our model has improved performance in MSE, PSNR, and SSIM. We also found that nearly 90% of the generated CTAs were of medium or high quality as assessed by clinicians. Upon comparing the synthesized CTA and authentic CTA assessments conducted by clinicians, our framework showed promising potential in diagnosing test sets on 426 paired CT-CTA images from three major medical centers, demonstrating improved results compared to the SOTA model.VesselTransGAN has the potential to improve the safety and accessibility of CTA scans for patients with renal insufficiency and to enhance the accuracy of CTA scans for diagnosing vascular disease. We release our code at https://github.com/Flora-huay/VesselTranGAN.

**INDEX TERMS** CTA, clinical diagnosis, generative model, image generation.

## I. INTRODUCTION

Computer Tomography Angiography (CTA) is a vascular imaging technique that uses a contrast medium to visualize the blood vessels. It can provide detailed information about the vascular anatomy and pathology, which is helpful for diagnosis and treatment planning, such as diagnosing coronary artery disease, vascular malformations, stenosis of the arteries in the head and neck, and assessing the risk of recurrent cerebral infarction [1].

However, injecting contrast medium poses risks to the body. The most serious risk associated with the use of iodinated contrast medium (ICM) is kidney injury, especially in patients with renal insufficiency [2]; their renal excretion will be delayed and further aggravate the damage to the whole body, such as allergic reactions [3] and thyrotoxicosis [4]. There is also evidence that contrast medium can cause DNA damage to blood cells, increase the risk of cancer, and even

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require dialysis treatment [5]. Therefore, reducing the use of contrast medium can benefit most clinical patients.

To overcome the challenges posed by contrast medium, various methods have been proposed to reduce the amount of contrast medium needed for CTA. One approach [6], [7] was to modify the concentration and injection rate of the iodine contrast medium, which could decrease the total amount of contrast medium needed. However, this might affect the image quality by introducing noise and artifacts. Another approach [8] was to optimize the image acquisition parameters of the CT scanner, such as voltage, pitch, collimation, and rotation times. This could enhance the image quality and reduce the contrast injection time by improving the temporal and spatial resolution. Johnson et al. [9] explored the initial experience of material differentiation using dual-energy CT, demonstrating the potential of this method in reducing the amount of contrast agent required. Carrascosa et al. [10] proposed a technique based on dual-energy single-energy reconstitution techniques and high-pitch techniques, which could minimize the contrast medium dose in CTA by utilizing the spectral properties of the X-rays and the high scanning speed. This technique could improve the contrast-to-noise ratio and mitigate the beam-hardening effect. Azzalini et al. [11] applied high-pitch spiral acquisition in a dual-source scanner, which could achieve ultra-fast imaging and lower the contrast medium exposure and display by 51%. This technique could diminish the motion artifacts and the radiation dose.

In recent years, deep learning models and self-attention mechanisms have made remarkable achievements in many fields, such as image processing. These techniques excel at capturing complex patterns and relationships in the data and mimicking how the human brain processes information. Jin et al. [12] leverages transfer learning and LSTM to analyze vehicle movement, with attention mechanisms highlighting key features. The EM algorithm refines filters for precise state estimation in challenging settings. Guo et al. [13]'s dynamic fusion strategy adapts to varying video frame features, enhancing video quality under different lighting conditions. CVANet [14] and GACNet [15] utilize deep networks and attention to improve image resolution. Li et al. [16] outlines retrieval models for efficient text-image feature integration. Dang and Lee [17]'s method fuses 2D and 3D data to enhance document image clarity. Addressing data scarcity, Dang and Lee [18] also introduces a GAN-based approach to generate paired image data from unpaired sources, processed by a U-net for improved data utility.

Inspired by the successful application of deep learning models in other fields, researchers began to apply deep learning techniques to medical image generation. This approach is expected to reduce the contrast medium required for CTA and improve the image quality, thereby providing accurate information for clinical diagnosis. For example, Chen et al. [19] developed a cascaded generative adjunctive network that synthesized 2D CTA slices from plain CT scans for aortic dissection diagnosis. Their method combined image registration, nnU-Net [20] segmentation, and a double discriminator network with a DCT channel attention mechanism. Lyu et al. [21] proposed a Generative Adversarial Network (GAN) based model that generated neck and abdominal CTA-like images without ICAs, achieving an SSIM of 0.906 on the external validation set. Killekar et al. [22] used a conditional GAN to generate pseudo-contrast CTA from perfectly registered non-contrast thin slice CT. Lyu focused on the production of large blood vessels in the neck and abdomen, and Killekar concentrated on the assessments of cardiac anatomy. Their methods could replace contrast CTA with pseudo-contrast CTA for some clinical applications, such as cardiac anatomy assessments. However, the production of small blood vessels in the brain was still ignored in this research. Zhang et al. [23] found that generating CTA images with ultra-low-dose angiographic contrastÂ agents by the MALAR method can significantly improve the clarity of blood vessels byÂ comparing images with standard images. Wu et al. [24] proposed using the VesselGan method to generate high-quality CTA images from myocardial perfusion CTP data, thereby reducing the use of contrast agents and the hazards associated with radiation.

Despite the advances in medical image generation using deep learning, many existing techniques suffered from low image quality and limited clinical applicability. To address these problems, we designed a VesselTranGAN model that generated CTA images focusing on small blood vessels in the brain from plain CT scans. Moreover, Our network consisted of two main components: a U-Net-like [25] generator with a self-attention mechanism and a 2D/3D data fusion strategy. Finally, we performed rigorous clinical evaluations to validate the utility and accuracy of our proposed method in real-world scenarios.

Our main contributions are:

(1) Enhanced Vascular Focus in GAN Network: We incorporated the Transformer [26] architecture into the GAN network and utilizes its self-attention mechanism to enhance focusing ability on the vascular regions. This improvement significantly improved the quality of the generated CTA images, especially in preserving the fine details and structures of the blood vessels.

(2) Exploration of 2D/3D Data Fusion: We compared three input strategies, including 3D data source, split 2D images, and 2D/3D fusion strategy. The results witnessed a significant improvement in 2D/3D fusion strategy.

(3) Clinical Evaluation and Validation: We conducted a comprehensive clinical evaluation to assess the practicality and reliability of our generated CTA images. We incorporated feedback from clinical experts and demonstrated the efficacy of our algorithm in producing clinically relevant and valuable outcomes.

In summary, the study aimed to generate CTA from CT scans by leveraging the Generative Adversarial Network (GAN) framework and 2D-3D fusion strategy. We proposed a GAN-based deep learning generative model that generated CTA images from head and neck CT scans. We adopted a U-Net-like generator to enhance the model's ability to focus on the vascular part by combining curvature and pixels. A 2D/3D data fusion method was additionally proposed to optimize the generation effect. Finally, clinical evaluations were performed to verify the practical value of our proposed method in real-world applications.

## II. METHODS

We proposed a GAN-based framework designed to generate CTA from standard CT scans. Our framework incorporated a self-attention mechanism into the generator, which enhances the capture of long-range dependencies and the target attention of blood vessel generation. Additionally, we utilized a fusion algorithm that combined the output of 3D and 2D models, leveraging the total performance in the generation task. The flowchart in 1 and 2 shows our CTA generation framework.

### A. VESSELTRANSGAN ARCHITECTURE

Our framework consists of two main components: 2D and 3D networks. The 2D network took CT slices as input and utilized a 2D convolutional network to generate 2D CTA images sequentially. The 3D network took CT patches (e.g. 32 layers) as input and used a 3D convolutional network to generate 3D CTA patches directly. The final CTA images were obtained through a fusion strategy by stacking 2D and 3D-generated CTA images. In addition to the fundamental GAN loss, our method incorporated pixel and vgg loss to improve the quality of generated vascular structures.

#### 1) NETWORK STRUCTURE

3 shows the architecture of our U-Net-like generator, which comprises two modules: an encoder and a decoder. The encoder generator had multiple sub-modules, each containing a conv2d encoder block, a down-sampling block, and an attention block. The decoder generator had similar sub-modules, each containing a decoder block, a down-sampling block, and an attention block. The encoding and decoding generators traversed linearly and had a feed-forward mechanism between internal sub-modules. The last four decoder sub-modules had an additional ToRGB layer to generate four CTA images at different resolutions. The transformer blocks were incorporated into both the encoder and the decoder, enhancing the network's ability to capture global information, which was beneficial for modeling intricate blood vessels.

Inspired by StyleGAN2 [27], our discriminator had a design with residual connections and down-sampling. It transformed the input image into a feature map of the exact resolution, passing through blocks with residual connections. Each block involved 2× down-sampling and two 3×3 convolutions with a central residual connection. This design enhanced the model's ability to capture intricate details, which requires local attention patterns for blood vessel generation. Residual connections facilitated effective learning and discrimination of local and global features, contributing to image authenticity evaluations.

#### 2) 2D-3D FUSION STRUCTURE

We propose a 2D-3D fusion strategy that combines the advantages of 2D slices and 3D reconstructions in medical imaging. The 2D slices process each CT layer independently to generate the corresponding CTA layer, capturing finer details but exhibiting poorer vascular continuity between adjacent layers. The 3D reconstructions take multiple CT layers as input to generate a 3D CTA image, providing coarser details but better vascular continuity between layers. Our study uses VesselTransGAN-2D and VesselTransGAN-3D models to produce 2D and 3D CTA images from CT scans, respectively. Through image alignment and normalization, we ensure the consistency of the image data. Moreover, we utilize Laplacian and Gaussian filters to capture the contour and detailed features of blood vessels, reduce the noise, and enhance the clarity of vascular structures, which is crucial for identifying and assessing vascular pathologies. By fusing the features of 2D and 3D images with a weighting algorithm, we can generate higher resolution and richer information images. Furthermore, these integrated images provide clinicians with a more comprehensive and in-depth view of the vascular structure, enhancing the accuracy of disease diagnosis and optimizing treatment plans. Specifically, we denoted the CTA generated by the 2D sub-model and the 3D sub-model as $C_{2d}$ and $C_{3d}$, respectively. Both $C_{2d}$ and $C_{3d}$ had dimensions $(H, W, D)$, where $H$, $W$ and $D$ represented the height, width and depth of a CTA slice. We performed the fusion process layer by layer. For simplicity, we assumed that $I_{2d}$ and $I_{3d}$ represent one layer each of $C_{2d}$ and $C_{3d}$ respectively, both having dimensions $(H, W)$. $I_{2d}$ and $I_{3d}$ were two frames perfectly aligned in position, maintaining the same indexing and dimensions $(H, W)$.

Initially, we normalized $I_{2d}$ and $I_{3d}$, ensuring that all the pixel were clipped within the range [0, 1]. To achieve this, we employed a box filter on each layer of $I_{2d}$ and $I_{3d}$, resulting in the base image $B_{2d}$ and $B_{3d}$, each having the same dimensions as $I_{2d}$ and $I_{3d}$. The formula for the box filter operation was given by:

$$B_{2d/3d}(x, y) = \frac{1}{r_{box}^2} \sum_{i=-r_{box}}^{r_{box}} \sum_{j=-r_{box}}^{r_{box}} I_{2d/3d}(x+i, y+j). \quad (1)$$

Subsequently, we calculated the detail images $D_{\mathrm{2d}}$ and $D_{\mathrm{3d}}$ by subtracting the base images from the original images. $D_{2d}$ and $D_{3d}$ shared the same dimensions as $I_{\mathrm{2d}}$ and $I_{\mathrm{3d}}$, represented as $(H, W, D)$. The calculation was expressed as:

$$D_{\mathrm{2d/3d}} = I_{\mathrm{2d/3d}} - B_{\mathrm{2d/3d}}. \quad (2)$$

Next, we applied Laplacian filtering to both input images to obtain high-frequency information.

$$H_{\mathrm{2d/3d}}(x, y) = |\nabla^2 I_{\mathrm{2d/3d}}(x, y)|. \quad (3)$$

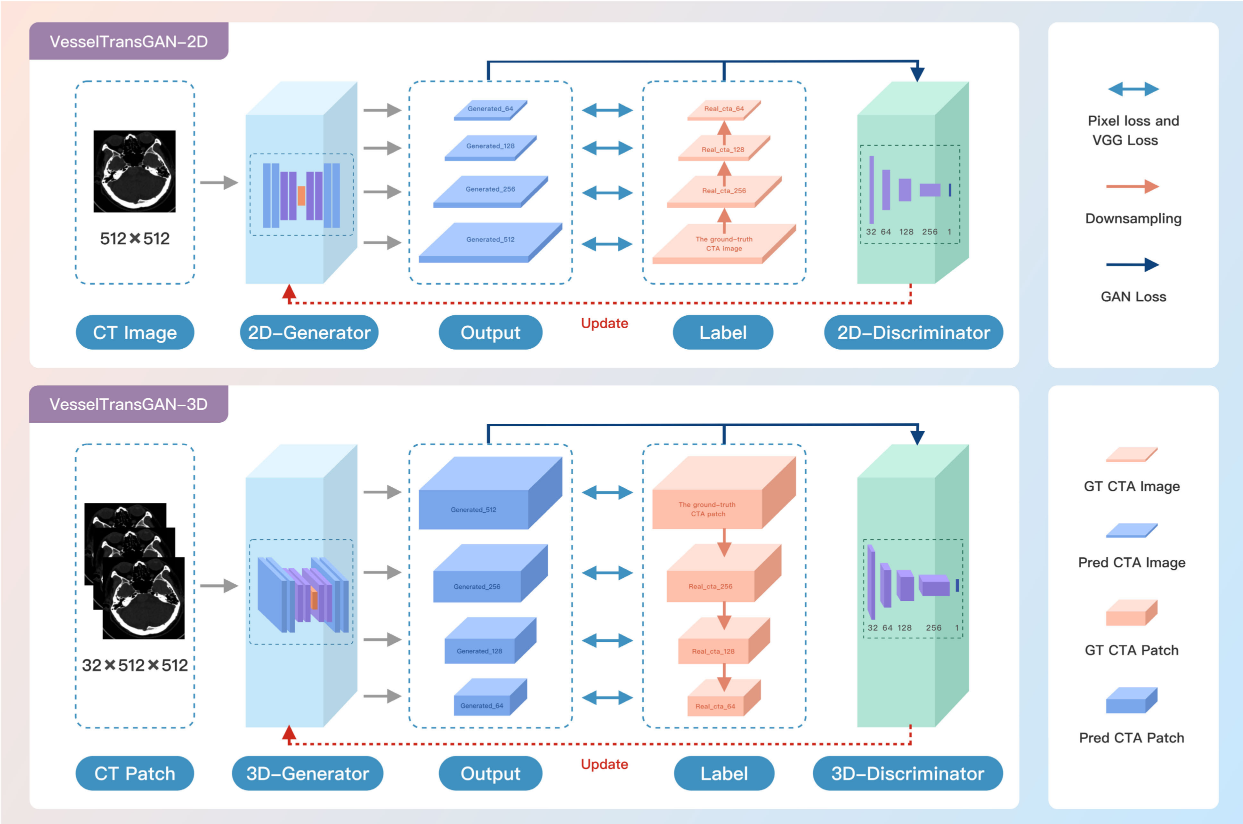


**FIGURE 1. Overview of Our GAN-based Framework. 1)The input of the VesselTransGAN-2D are 512 × 512 2D CT images. The input of the 3D-Generator is bx512 × 512 3D CT Patches. 2)The output of the 2D-Generator is four generated CTA images with different resolutions. The output of the 3D-Generator is four generated CTA patches with different resolutions. 3)The Label contains one ground-truth CTA image and three low-resolution CTA images, with downsampling from the ground truth. 4)The 2D-Discriminator can calculate the GAN loss using the generated CTA image and Label. The 3D-Discriminator can calculate the GAN loss using the generated CTA patch and Label.**

We then performed Gaussian filtering on the high-frequency information to smooth out noise.

$$S_{2d/3d}(x,y) = \frac{1}{2\pi\sigma_{xy}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{xy}^2}\right) * H_{2d/3d}(x,y). \quad (4)$$

Here, $S$ represented the Gaussian filtering, and $\sigma_{xy}$ represented the covariance of x and y. Next, we calculated the weight mappings $w_{2d}$ and $w_{3d}$ by determining the image with higher pixel values at each pixel position through a comparison of the pixel values in the two images.

$$w_{2d} := \begin{cases} 1 & \text{if } I_{2d}(x,y) = \max(I_{2d}(x,y), I_{3d}(x,y)). \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

$$w_{3d} := 1 - w_{2d} \quad (6)$$

We created guided filters with different radius parameters, using img1 and img2 as guide images. This led to the generation of four guided filters: $f_{\text{base2d}}$, $f_{\text{base3d}}$, $f_{\text{detail2d}}$, and $f_{\text{detail3d}}$. Subsequently, we applied these filters to the previously obtained $w_{2d}$ and $w_{3d}$, where $w_{\text{base2d}} = f_{\text{base2d}}(w_{2d})$, $w_{\text{base3d}} = f_{\text{base3d}}(w_{3d})$, $w_{\text{detail2d}} = f_{\text{detail2d}}(w_{2d})$, $w_{\text{detail }3d} = f_{\text{detail3d}}(w_{3d})$.

In these expressions, $w_{\text{base2d}}$ and $w_{\text{base3d}}$ represented the base components, while $w_{\text{detail2d}}$ and $w_{\text{detail3d}}$ represented the detail components of the corresponding guided filters applied to the weight maps $w_{2d}$ and $w_{3d}$.

Finally, we obtained the final fused image by performing a weighted sum of the base and detail layers based on the normalized weights.

$$\text{BASE} = B_{2d} \times w_{\text{base2d}} + B_{3d} \times w_{\text{base3d}}. \quad (7)$$

$$\text{DETAIL} = D_{2d} \times w_{\text{detail2d}} + D_{3d} \times w_{\text{detail3d}}. \quad (8)$$

This fusion strategy allowed us to achieve higher-quality CTA images by combining the advantages of both 2D and 3D models.

## III. EXPERIMENTS AND RESULTS

### A. DATA ACQUISITION AND PROCESSING

Figure 4 showed our project's data collection and processing pipeline. We initially collected 14,880 patients with brain/neck CT or CTA scans. We enrolled 2610 pairs of unenhanced CT and unenhanced thin-slice CTA data with matching numbers and positions through manual screening. The CT plain scan and CTA data were precisely aligned, ensuring that CTA data could serve as labels to supervise the generated CTA quality. Each pair of CT and CTA with a less than 5 mm thickness was resampled to 0.537mm ×

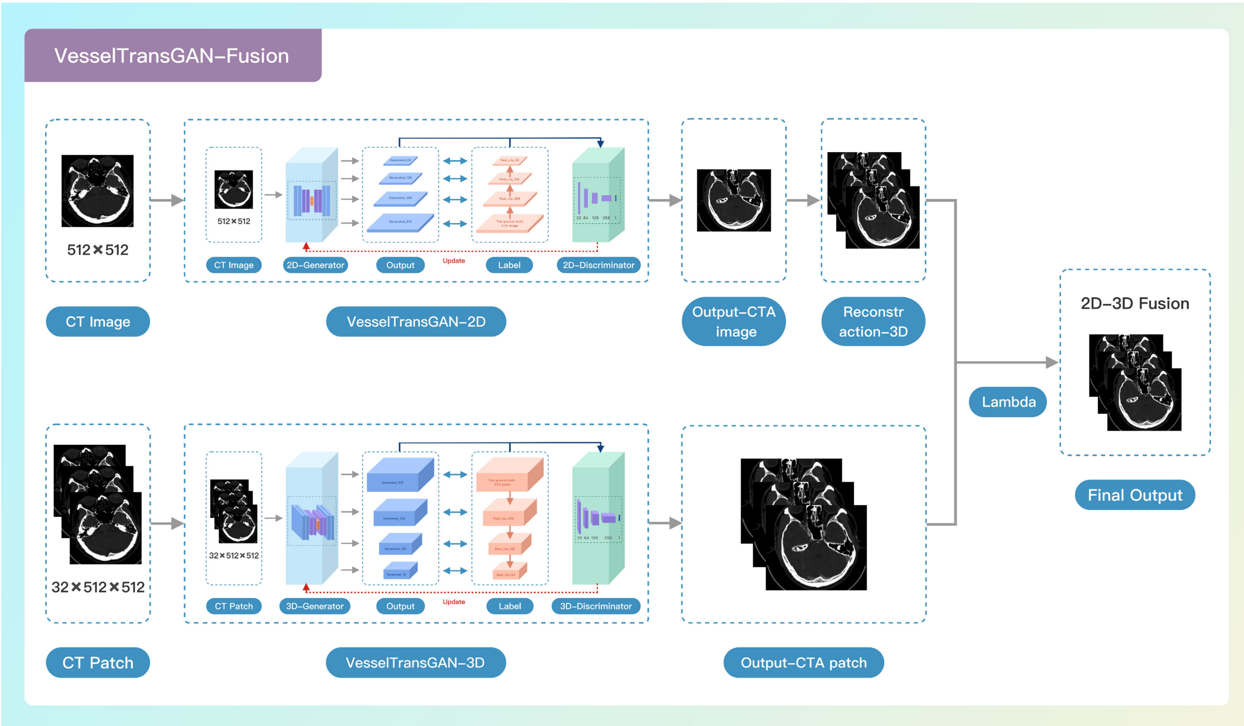


**FIGURE 2. VesselTransGAN-Fusion: A 2D-3D Fusion for CT Image Reconstruction. 1) Generate a CTA image through VesselTransGAN-2D model and reconstruct it into the 3D patch. 2) Generate CTA patches through the VesselTransGAN-3D model. 3) Weighted fusion of 2D and 3D image features by Lamda coefficient. 4) Generate a 2D-3D fusion patch. 2D model provides detailed learning for individual CT slices, while the 3D model ensures better continuity in vascular structures due to its holistic approach. Our 2D-3D fusion strategy optimizes both slice detail and vascular continuity, combining the best of both modeling techniques.**

0.537mm × 0.625mm volumes, consisting of 100-650 sections with a matrix size of 512 × 512 pixels. The dataset was randomly split into train, validation, and test sets with a ratio of 7:1.5:1.5.

The contrast medium utilized for all CTA examinations was iodine, with a concentration of 370 mg/mL and an injection rate of 4.5 mL/sec. By utilizing pixel grayscale alignment, we can significantly improve the local visualization quality of blood vessels. This method can reduce image noise and enhance the clarity of blood vessel edges, making the contours of the vessels clearer, thus providing clinicians with more precise vascular images. This is essential for clinical applications such as early diagnosis of vascular diseases, diagnostic decisions, assessment of treatment effects, and planning of surgical pathways to increase the recovery rate of patients. Each volume underwent independent processing, normalizing intensities from -2000 to 2095 pixel value to -1 to 1. Scans with poor image quality upon manual inspection were excluded based on the defined inclusion and exclusion criteria. The following list outlines the different types of graphics published in IEEE journals. They are categorized based on their construction and use of color/shades of gray:

Our final dataset comprised 2610 pairs of aligned CT plain scans and CTA data. We allocated 1801 paired CT-CTA images for training and the remaining for validation and testing. The test set contained 426 paired CT-CTA images, of which 389 were original center data, and 37 were external validation data. External validation data included 16 cases from the Sixth Medical Center of the PLA General Hospital and 21 cases from Beijing Tiantan Hospital. In clinical applications, CTA scans are conducted on patients with various diseases. To ensure the model's generalizability, our dataset involves a variety of cerebrovascular diseases, such as intracranial aneurysms, cerebral artery atherosclerosis, and arterial dissections.

### B. TRAINING AND IMPLEMENTATION DETAILS

Our training strategy employs LSGAN for adversarial training. The loss function used in adversarial training is defined as

$$L_{\mathrm{GAN}} = \mathrm{Mean}(\mathrm{D}(\mathrm{CTA})\text{-}1)^2 + \mathrm{Mean}(\mathrm{D}(\mathrm{G}(\mathrm{CT})))^2 \qquad (9)$$

where D(real_samps) represented predictions of the discriminator for real samples, D (G(fake_samps)) represented predictions of the discriminator for fake samples. Mean(·) calculated the mean of the predictions. The target for real samples is 1, and for fake samples, it is 0. The loss was computed as the mean squared difference between predictions and the target values.

For utilizing multi-resolution outputs for generator supervision, VGG loss was applied to the generator's multi-resolution outputs for high-dimensional feature supervision, and L1 loss was used for per-pixel supervision of the generator's outputs. This comprehensive training approach

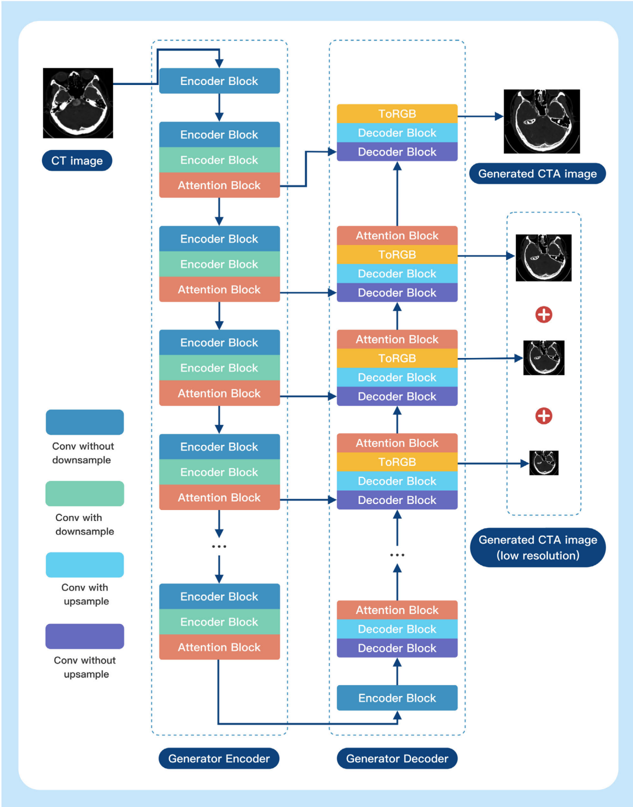


**FIGURE 3.** The Architecture of Generator. 1) The generator has a U-Net-like architecture, and the input CT image is first sent into the 2D/3D generator encoder. 2) Each encoder layer consists of two encoder blocks and an attention block. The first encoder blocks convs without downsample, and the second is convs with downsample. 3) The downsample helps to generate different fake CTA images with different resolutions. The attention block enhances the critical features while suppressing the less informative ones.4) Each decoder layer consists of two decoder blocks and an attention block. The first decoder block convs without upsample, and the second decoder block is convs with upsample. The ToRGB layer is used to generate fake CTA images.

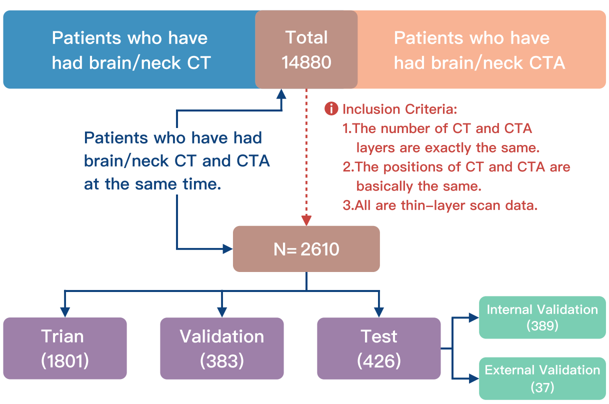


**FIGURE 4.** The overview of data acquisition and processing.

aimed to ensure accurate and detailed synthesis of 3D medical images.

The VGG loss leveraged a pre-trained VGG model to compare high-level features of the generator's output and the target image. It provided high-dimensional feature supervision that could capture structural and content differences. The resize operation ensured that both images were of comparable size before computing the loss.

This study used Python version 3.11.4 for all experiments and analyses. Our implementation leveraged the Distributed Data Parallelism (DDP) capabilities provided by the widely adopted open-source deep learning framework PyTorch, version 2.1.2, with CUDA 12.2 support. The model training was conducted over 61 hours on four NVIDIA A40 GPUs, each equipped with 48 GB of memory.

### C. EVALUATION

#### 1) QUANTITATIVE EVALUATION

The results of the proposed GAN model were compared quantitatively with those of three other image translation models: Pix2Pix [28], CycleGAN [29],and ACL-GAN [30]. Mean Square Error (MSE), peak signal-to-noise ratio (PSNR), and the structural similarity index measure (SSIM) [31] were used to evaluate the quality of images produced by real CTA and generated CTA.

**TABLE 1.** The evaluation metrics for model comparison. (The bold numbers represent the highest one of each indicator).

| Method(CT→CTA) | MSE↓ | PSNR↑ | SSIM↑ |
|---|---|---|---|
| Pix2Pix [28] | 56.66 | 31.16 | 0.577 |
| CycleGAN [29] | 12.35 | 37.97 | 0.903 |
| ACL-GAN [30] | 60.54 | 30.45 | 0.680 |
| VesselTransGAN-2D | **10.09** | **38.37** | 0.926 |
| VesselTransGAN-3D | 10.77 | 38.05 | 0.926 |
| VesselTransGAN-Fusion | 10.36 | 38.22 | **0.929** |

From the table 1, we observe that Pix2Pix exhibits the highest MSE and the lowest SSIM, which indicates that it struggles with maintaining the structural integrity of the images and introduces more artifacts. CycleGAN improves significantly over Pix2Pix, particularly in terms of SSIM and PSNR, demonstrating its ability to better preserve structural and visual quality through unpaired image-to-image translation. However, ACL-GAN, despite its popularity, shows a notable decline in performance with an MSE of 60.54, a PSNR of 30.45, and an SSIM of 0.68. These results suggest that ACL-GAN is less effective for the CT to CTA conversion task, potentially due to its difficulty preserving critical image features and maintaining high image quality. Our VesselTransGAN models outperform these baselines. Specifically, VesselTransGAN-2D achieves the lowest MSE and highest PSNR, indicating that it excels in reducing reconstruction errors while maintaining high image quality. VesselTransGAN-3D also performs admirably, particularly in SSIM, reflecting its strength in preserving structural similarity across slices of volumetric data.

Finally, VesselTransGAN-Fusion strikes a balance across all metrics, with an MSE lower than most baselines, a high PSNR, and the highest SSIM. This suggests that the fusion approach effectively combines the strengths of both 2D and

3D data processing, resulting in superior image quality and structural preservation. The quantitative results demonstrate that our VesselTransGAN models, particularly the fusion variant, provide a more robust and accurate solution for CT to CTA conversion compared to traditional methods.

As a form of expression commonly used in signal processing, PSNR is not fundamentally different from MSE. The problem with MSE and PSNR is that when calculating the differences in pixels at each location, their results are only related to the two-pixel values at the specific position rather than the pixels at any other location. That is to say, these two methods of calculating only treat the image as isolated pixels and ignore some of the visual features contained in the image content, especially the local structural information of the image. To a large extent, image quality is a subjective feeling in which the structural information significantly impacts people's subjective feelings. SSIM was proposed to solve the problem of MSE and PSNR mentioned above, and when SSIM calculates the differences between two images at each position, instead of taking one pixel from each of the two images at that location, one pixel from each area is taken. Our VesselTransGAN-Fusion, combining 2D and 3D, performed well, achieving an SSIM value of 0.929.

Figure 5 listed a visual comparison of models. Compared to the CycleGAN, Pix2Pix, and ACL-GAN models, our model provided a more precise depiction of the size and morphology of intracranial blood vessels.

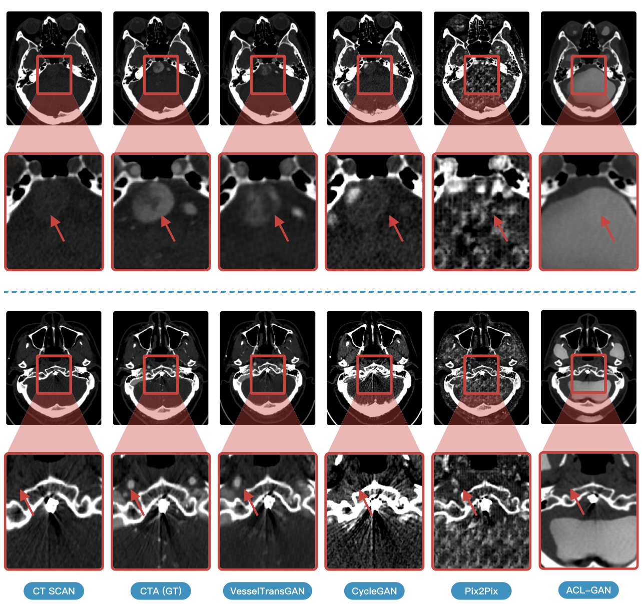


**FIGURE 5.** **Visualization. The pictures above were two visualization examples; the second row was a local enlargement of the first. The original CTA image revealed a fusiform dilation in the intracranial segment of the right vertebral artery, extending to the proximal segment of the basilar artery with an intramural thrombus. Compared to the CycleGAN, Pix2Pix, and ACL-GAN models, our model provided a clearer depiction of the size and morphology of the arterial aneurysm and the intramural thrombus. In the fourth row, our model displayed the intracranial small vessels relatively clearly, even with radial metallic artifacts, while the comparison models failed to observe the intracranial vessels.**

**TABLE 2.** **The Evaluation metrics for ablation experiment. (The bold numbers represent the highest one of each indicator).**

| Method(CT→CTA) | MSE↓ | PSNR↑ | SSIM↑ |
|---|---|---|---|
| VesselTransGAN-Fusion | **10.36** | **38.22** | **0.929** |
| -No our Self-Attention | 13.42 | 37 | 0.913 |

As demonstrated by Table 2, the removal of the attention block resulted in a significant increase in MSE and a notable decrease in both PSNR and SSIM for the VesselTransGAN-Fusion model. This underscores the pivotal role of the attention mechanism in enhancing model performance and image quality.

Qualitatively, we observed that the self-attention mechanism helps to enhance the overall performance of the model, particularly in preserving finer details and improving image reconstruction quality. For example, the MSE (Mean Squared Error) is significantly reduced from 13.42 to 10.36 when self-attention is included, indicating a more accurate reconstruction. Similarly, the PSNR (Peak Signal-to-Noise Ratio) is improved from 37.00 to 38.22, suggesting better image quality with less noise. The SSIM (Structural Similarity Index Measure) also shows a noticeable improvement from 0.913 to 0.929, reflecting that the images generated with self-attention are more structurally similar to the ground truth. The quantitative results indicate that the addition of the self-attention module in VesselTransGAN-Fusion enhances the model's capability to focus on important features, leading to better overall image quality across all evaluated metrics.

### 2) HUMAN EVALUATION

200 patients' CTA images were evaluated, 100 of which were generated by our model, and another 100 were real CTAs. Randomly mixed CTA images were independently handed over to two radiologists (with 5 and 8 years of diagnostic experience), unaware of the data source (real or generated), to assess the image quality of craniocerebral CTA.

**TABLE 3.** **The quality evaluation of real and generated CTA Image.**

| | Poor Quality | Medium Quality | High Quality |
|---|---|---|---|
| doctor 1 | 10% | 10% | 80% |
| doctor 2 | 8% | 11% | 81% |

Cerebrovascular radiography experts utilized a subjective three-point scale (visual quality score) for image quality assessment, with scores ranging from 1 (poor quality) to 3 (good quality). Specifically, the evaluation encompassed subcategories such as vessel wall clarity, vessel continuity, vessel boundary, small vessel clarity, intracranial soft tissue, and artifact [32], [33]. When assessing the quality of the generated intracranial CTA, our method achieved better results.

We evaluated the quality of those 100 generated CTA images by statistically analyzing the subjective scoring of radiologists. The quality evaluation result is shown in Table 3.

It showed that at most 10% of the results generated by the VesselTransGAN model were rated as poor quality, and more than 90% were rated as medium or high quality.

### 3) DIAGNOSIS EVALUATION

The quality of CTA images could affect radiologists' diagnosis and decision-making regarding vascular diseases, making subjective assessments of visualization quality crucial. Two experienced radiologists evaluated real CTA images and diagnosed intracranial aneurysms, cerebral artery sclerosis, and arterial dissections. The vascular diagnoses obtained from this step were considered the clinical standard and denoted as $R$.

**TABLE 4.** **The diagnosis evaluation metric of two diseases: Clinician expert based on generated CTA.**

| Disease | Precision | Recall | F1 score |
|---|---|---|---|
| Intracranial Aneurysm | 0.875 | 0.636 | 0.737 |
| Cerebral Arteriosclerosis | 0.860 | 0.961 | 0.908 |

To ensure objectivity and fairness in subjective assessments, two additional senior radiologists independently evaluated the quality of synthesized CTA images and derived vascular diagnoses, denoted as $R'$.

We compared $R$ and $R'$ to discern the value of synthesized CTA in the diagnostic process. Precision, Recall, and F1 scores were used to evaluate the model's output for each diagnosis category and overall accuracy. The F1 score metric was used to assess the overall performance of the CTA-GAN model.

Two radiologists diagnosed the generated 100 CTA models. Compared with the existing real CTA diagnosis results, the 10% data with poor generation effect were eliminated, and T was calculated to obtain Table 4. It could be seen that the diagnostic accuracy of our model for Intracranial Aneurysm reached 0.875, the recall rate reached 0.636, the F1 value was 0.737, and the diagnostic accuracy of Cerebral Arteriosclerosis reached 0.860, the recall rate was 0.961, and the F1 value reached 0.908, which demonstrated a good clinical diagnostic result.

## IV. CONCLUSION

Cerebral computed tomography angiography (CTA) is widely used to diagnose vascular pathologies, which requires the application of a contrast medium for precise arterial visualization. Alternative diagnostic methods, such as magnetic resonance angiography (MRA), are recommended for patients with contraindications to contrast medium. However, these alternative therapies are usually hard to afford and time-consuming for patients [34].

In response to this challenge, our research introduces 'VesselTransGAN,' a GAN-based model that generates CTA images from standard CT scans without needing the contrast medium. Thus, it significantly lowers the risk factor of patients unable to undergo traditional contrast-enhanced CTA. By incorporating a self-attention mechanism and a 2D-3D fusion strategy, this model not only improves the clarity of vascular visualization but also extends the benefits of CTA imaging to a broader range of patients, including those with contraindications to contrast medium.

Despite these advancements, our study has some limitations. Firstly, Some clinicians have indicated that the generated images may lack perfect continuity of blood vessels, potentially leading to misdiagnoses of conditions like vascular occlusion. Moreover, the model's performance may be constrained by training time, computational resources, and data quantity and quality. Finally, Although the images generated by the GAN technology employed in this study exhibit a high degree of fidelity to real images, they still fall short of the stringent criteria required for clinical use. Further research is necessary to ensure they adhere to the rigorous standards of accuracy, consistency, and diagnostic relevance.

To address these limitations and further refine our model, we have outlined several future directions. Firstly, We plan to initiate the fusion process earlier in the network rather than waiting until the final generation stage. Zhao et al. [35] regarded CNN and transformer branches as two student models and got an information fusion by averaging the outputs of the two branches [35]. Hussain et al. [36]proposed a character-level CNN that could extract multi-level features to identify malicious and benign URLs. Li et al. [37] fused feature maps from multiple stages in weakly supervised pathological image segmentation tasks, achieving better results than single-stage fusion. We believe this modification will significantly improve the accuracy and visual quality of the generated CTA scans. Secondly, as generating small blood vessels is crucial in our task, we plan to focus on it in future work. For example, Liu et al. [38] proposed an energy-based loss function in 3D segmentation to enhance the continuity of blood vessels. Thirdly, we plan to optimize the network architecture and develop new loss functions to capture the continuity of the generated images better. Finally, we will extend our research to include a comparative analysis of various attention mechanisms, such as global, local, multi-head, spatial, and channel attention, which will provide a deeper understanding of their impact on vascular segmentation and overall image quality in medical imaging.

## DECLARATION OF INTERESTS

The authors declare no competing interests. Their patent related to this work was China Patent 2022-766-01.

## REFERENCES

[1] W. Brant, *The Brant and Helms Solution: Fundamentals of Diagnostic Radiology*. Philadelphia, PA, USA: Lippincott Williams & Wilkins, 2006.
[2] E. Matthews, "Acute kidney injury and iodinated contrast media," *Radiol. Technol.*, vol. 89, no. 5, pp. 467–477, 2018.
[3] Y.-W. Wu, K. S. Leow, Y. Zhu, and C. H. Tan, "Prevention and management of adverse reactions induced by iodinated contrast media," *Ann. Acad. Med., Singap.*, vol. 45, no. 4, pp. 157–164, Apr. 2016.
[4] A. Rusandu, B. H. Sjøvold, E. Hofstad, and R. J. Reidunsdatter, "Iodinated contrast media and their effect on thyroid function—Routines and practices among diagnostic imaging departments in Norway," *J. Med. Radiat. Sci.*, vol. 67, no. 2, pp. 111–118, Jun. 2020.

[5] R. Harbron, E. A. Ainsbury, S. D. Bouffler, R. J. Tanner, J. S. Eakins, and M. S. Pearce, "Enhanced radiation dose and DNA damage associated with iodinated contrast media in diagnostic X-ray imaging," *Brit. J. Radiol.*, vol. 90, no. 1079, pp. 1–3, Nov. 2017.

[6] Y. Ozawa, M. Hara, and Y. Shibamoto, "The frequency of insufficient contrast enhancement of the pulmonary artery in routine contrast-enhanced chest CT and its improvement with an increased injection rate: A prospective study," *J. Thoracic Imag.*, vol. 26, no. 1, pp. 42–47, 2011.

[7] H. Alobeidi, M. Alshamari, J. Widell, T. Eriksson, and M. Lidén, "Minimizing contrast media dose in CT pulmonary angiography with high-pitch technique," *Brit. J. Radiol.*, vol. 93, no. 1111, Jul. 2020, Art. no. 20190995.

[8] K. Kaproth-Joslin, S. Hobbs, P. Rajiah, A. Chaturvedi, and A. Chaturvedi, "Optimizing low contrast volume thoracic CT angiography: From the basics to the advanced," *J. Clin. Imag. Sci.*, vol. 12, p. 41, Jul. 2022.

[9] T. R. C. Johnson, B. Krauß, M. Sedlmair, M. Grasruck, H. Bruder, D. Morhard, C. Fink, S. Weckbach, M. Lenhard, B. Schmidt, T. Flohr, M. F. Reiser, and C. R. Becker, "Material differentiation by dual energy CT: Initial experience," *Eur. Radiol.*, vol. 17, no. 6, pp. 1510–1517, May 2007.

[10] P. Carrascosa, C. Capunay, G. A. Rodriguez-Granillo, A. Deviggiano, J. Vallejos, and J. A. Leipsic, "Substantial iodine volume load reduction in CT angiography with dual-energy imaging: Insights from a pilot randomized study," *Int. J. Cardiovascular Imag.*, vol. 30, no. 8, pp. 1613–1620, Dec. 2014.

[11] L. Azzalini, S. Abbara, and B. B. Ghoshhajra, "Ultra-low contrast computed tomographic angiography (CTA) with 20-mL total dose for transcatheter aortic valve implantation (TAVI) planning," *J. Comput. Assist. Tomogr.*, vol. 38, no. 1, pp. 105–109, 2014.

[12] X. Jin, T. Sun, W. Chen, H. Ma, Y. Wang, and Y. Zheng, "Parameter adaptive non-model-based state estimation combining attention mechanism and LSTM," *IECE Trans. Intell. Systematics*, vol. 1, no. 1, pp. 40–48, May 2024.

[13] X. Guo, F. Yang, and L. Ji, "A mimic fusion algorithm for dual channel video based on possibility distribution synthesis theory," *Chin. J. Inf. Fusion*, vol. 1, no. 1, pp. 33–49, May 2024.

[14] W. Zhang, W. Zhao, J. Li, P. Zhuang, H. Sun, Y. Xu, and C. Li, "CVANet: Cascaded visual attention network for single image super-resolution," *Neural Netw.*, vol. 170, pp. 622–634, Feb. 2024.

[15] W. Zhang, Z. Li, G. Li, P. Zhuang, G. Hou, Q. Zhang, and C. Li, "GACNet: Generate adversarial-driven cross-aware network for hyperspectral wheat variety identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503314.

[16] T. Li, L. Kong, X. Yang, B. Wang, and J. Xu, "Bridging modalities: A survey of cross-modal image-text retrieval," *Chin. J. Inf. Fusion*, vol. 1, no. 1, pp. 79–92, Jun. 2024.

[17] Q.-V. Dang and G.-S. Lee, "Utilizing 3D information from point clouds to support document image binarization," EasyChair Preprint, Chonnam Nat. Univ., Gwangju, South Korea, Tech. Rep. 9141, Oct. 2022.

[18] Q.-V. Dang and G.-S. Lee, "Document image binarization by GAN with unpaired data training," *Int. J. Contents*, vol. 16, no. 2, pp. 8–18, 2020.

[19] H. Chen, S. Yan, M. Xie, and J. Huang, "Application of cascaded GAN based on CT scan in the diagnosis of aortic dissection," *Comput. Methods Programs Biomed.*, vol. 226, Nov. 2022, Art. no. 107130.

[20] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, no. 2, pp. 203–211, Feb. 2021.

[21] J. Lyu, Y. Fu, M. Yang, Y. Xiong, Q. Duan, C. Duan, X. Wang, X. Xing, D. Zhang, J. Lin, C. Luo, X. Ma, X. Bian, J. Hu, C. Li, J. Huang, W. Zhang, Y. Zhang, S. Su, and X. Lou, "Generative adversarial network–based noncontrast CT angiography for aorta and carotid arteries," *Radiology*, vol. 309, no. 2, Nov. 2023, Art. no. e230681.

[22] A. Killekar, J. Kwiecinski, M. Kruk, C. Kepka, A. Shanbhag, and D. Dey, "Pseudo-contrast cardiac CT angiography derived from non-contrast CT using conditional generative adversarial networks," *Proc. SPIE*, vol. 12464, pp. 779–785, Apr. 2023.

[23] W. Zhang, Z. Zhou, Z. Gao, G. Yang, L. Xu, W. Wu, and H. Zhang, "Multiple adversarial learning based angiography reconstruction for ultra-low-dose contrast medium CT," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 1, pp. 409–420, Jan. 2023, doi: 10.1109/JBHI.2022.3213595.

[24] C. Wu, H. Zhang, J. Chen, Z. Gao, P. Zhang, K. Muhammad, and J. D. Ser, "Vessel-GAN: Angiographic reconstructions from myocardial CT perfusion with explainable generative adversarial networks," *Future Gener. Comput. Syst.*, vol. 130, pp. 128–139, May 2022.

[25] O. Ronneberger and P. Fischer, "Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[27] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of StyleGAN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8107–8116.

[28] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1125–1134.

[29] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[30] Y. Zhao, R. Wu, and H. Dong, "Unpaired image-to-image translation using adversarial consistency loss," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 800–815.

[31] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[32] X. Cui, W. Shi, K. Zhang, Y. Miao, and Z. Jiang, "Medical image quality assessment method based on residual learning," in *Proc. IEEE 5th Int. Conf. Signal Image Process. (ICSIP)*, Oct. 2020, pp. 195–200.

[33] X. Dai, M. Yu, J. Pan, Z. Lu, C. Shen, Y. Wang, B. Lu, and J. Zhang, "Image quality and diagnostic accuracy of coronary CT angiography derived from low-dose dynamic CT myocardial perfusion: A feasibility study with comparison to invasive coronary angiography," *Eur. Radiol.*, vol. 29, no. 8, pp. 4349–4356, Aug. 2019.

[34] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. Van Der Laak, B. Van Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Med. Image Anal.*, vol. 42, pp. 60–88, Dec. 2017.

[35] Z. Duan, X. Luo, and T. Zhang, "Combining transformers with CNN for multi-focus image fusion," *Expert Syst. Appl.*, vol. 235, Jan. 2024, Art. no. 121156.

[36] M. Hussain, C. Cheng, R. Xu, and M. Afzal, "CNN-fusion: An effective and lightweight phishing detection method based on multi-variant ConvNet," *Inf. Sci.*, vol. 631, pp. 328–345, Jun. 2023.

[37] K. Li, Z. Qian, Y. Han, E. I.-C. Chang, B. Wei, M. Lai, J. Liao, Y. Fan, and Y. Xu, "Weakly supervised histopathology image segmentation with self-attention," *Med. Image Anal.*, vol. 86, May 2023, Art. no. 102791.

[38] P. Liu, G. Huang, J. Jing, S. Bian, L. Cheng, X. Y. Lu, C. Rao, Y. Liu, Y. Hua, Y. Wang, and K. He, "An energy matching vessel segmentation framework in 3-D medical images," *IEEE Trans. Med. Imag.*, vol. 43, no. 4, pp. 1476–1488, Apr. 2024.


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